PROGRAM: // program for merge sort

```
#include <stdio.h>
#include <stdlib.h>

void merge(int arr[], int l, int m, int r){
   int i, j, k;
   int n1 = m - l + 1;
   int n2 = r - m;

   // Create temp arrays
   int L[n1], R[n2];

   // Copy data to temp arrays L[] and R[]
   for (i = 0; i < n1; i++)
      L[i] = arr[l + i];
   for (j = 0; j < n2; j++)
      R[j] = arr[m + 1 + j];

   // Merge the temp arrays back into arr[l..r
   i = 0;
   j = 0;
   k = l;
   while (i < n1 && j < n2) {
      if (L[i] <= R[j]) {
         arr[k] = L[i];
         i++;
      }
      else {
         arr[k] = R[j];
         j++;
      }
      k++;
   }
   // Copy the remaining elements of L[],  if there are any
   while (i < n1) {
      arr[k] = L[i];
      i++;
      k++;
   }
   // Copy the remaining elements of R[],   // if there are any
   while (j < n2) {
      arr[k] = R[j];
      j++;
      k++;
```

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   }
}

void mergeSort(int arr[], int l, int r){

   if (l < r) {
      int m = l + (r - l) / 2;

      // Sort first and second halves
      mergeSort(arr, l, m);
      mergeSort(arr, m + 1, r);

      merge(arr, l, m, r);
   }
}

// Driver code
int main(){

   int arr[] = {38, 27, 43, 10};
   int arr_size = sizeof(arr) / sizeof(arr[0]);
   int i;

      printf("before sorting \n");
   for (i = 0; i < arr_size; i++)
      printf("%d ", arr[i]);
   printf("\n");


   mergeSort(arr, 0, arr_size - 1);

      printf("before sorting \n");
   for (i = 0; i < arr_size; i++)
      printf("%d ", arr[i]);
   printf("\n");

   return 0;
}
```